

```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[GH_FFCutsFits,CutFitFileGH] =
recover_selected_cuts('GH_FFCutsFits','e:\SHARE\Fe\Data\analysis\07_fit_with_multicu
t_all\sel_cuts');
```

```
fn = fieldnames(GH_FFCutsFits);
%nf = 11
%for nf = 2:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei800_off110GHmaxFF';
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('*****\n');
```

```
*****
```

```
fprintf('Processing cuts: %s \n', selectCuts);
```

```
Processing cuts: Ei800_off110GHmaxFF
```

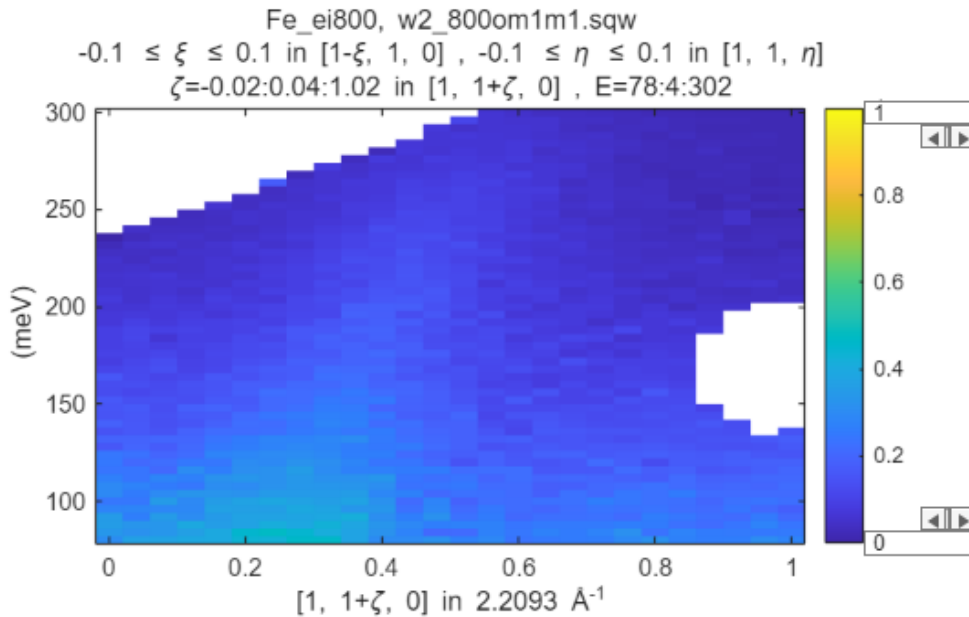
```
fprintf('*****\n');
```

```
*****
```

```
cuts_list = GH_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "GH";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

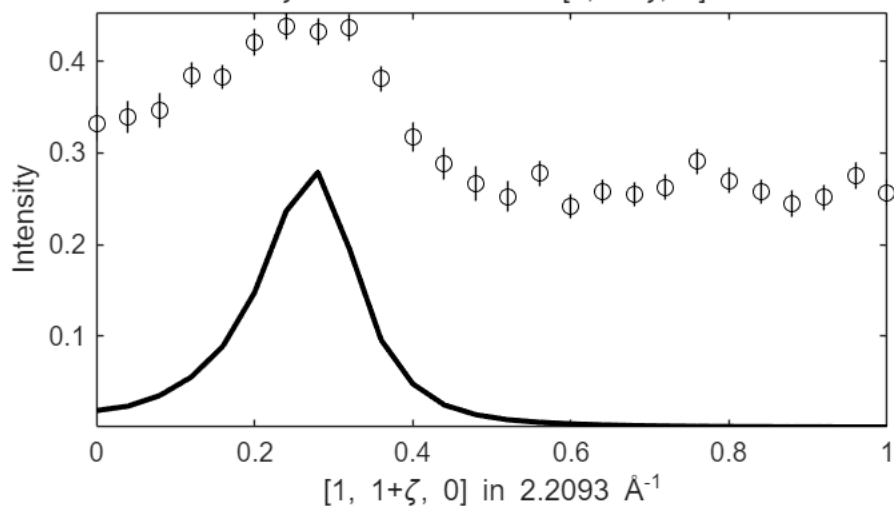
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GH_FFCutsFits.(selectCuts) = cuts_list;

GH_FFCutsFits = fit_multicutsFF_along_direction(...
    GH_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

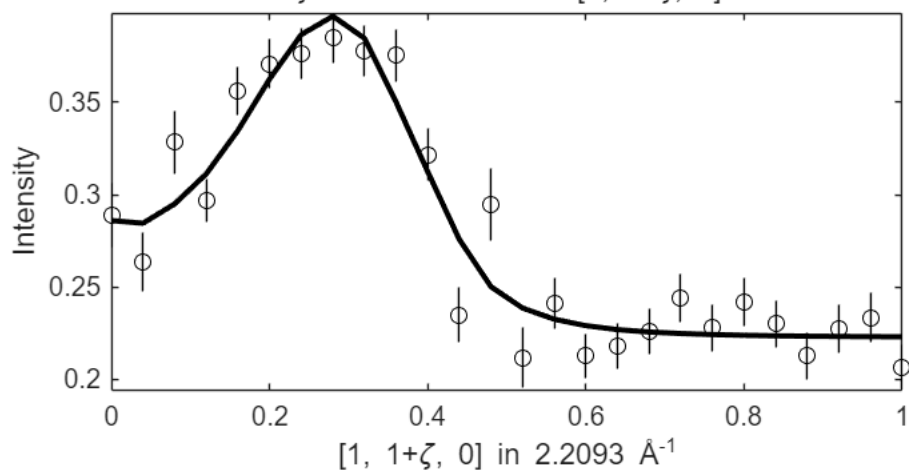
```
*****
**** En = 83±5 going to 293
*****
```

Fe_ei800, w2_800om1m1.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $78 \leq E \leq 88$ eV
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$

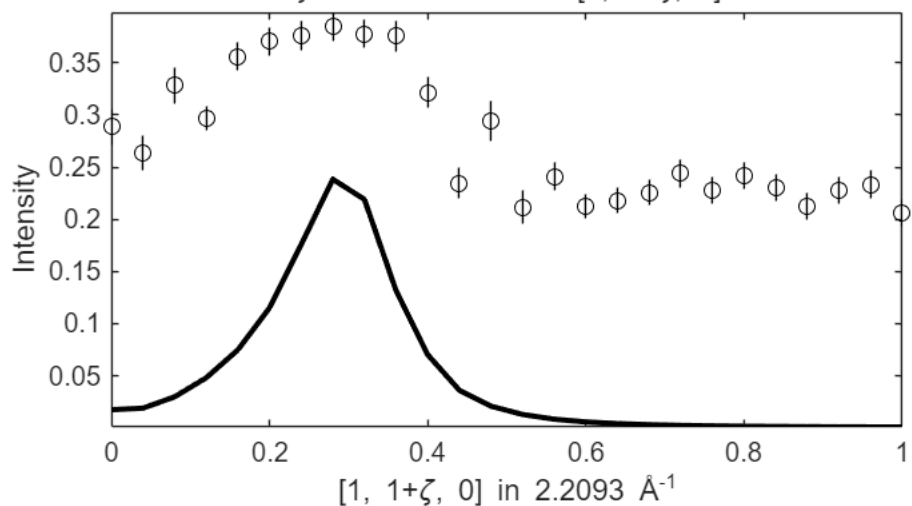


 **** En = 93±5 going to 293

Fe_ei800, w2_800om1m1.sqw
 $S=0.95 \pm 0.046$; $J_0 = 31 \pm 1.3$; $J_1 = 0 \pm 0$; $\gamma = 52.5139 \pm 2.09425$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $88 \leq E \leq 93$ eV
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$

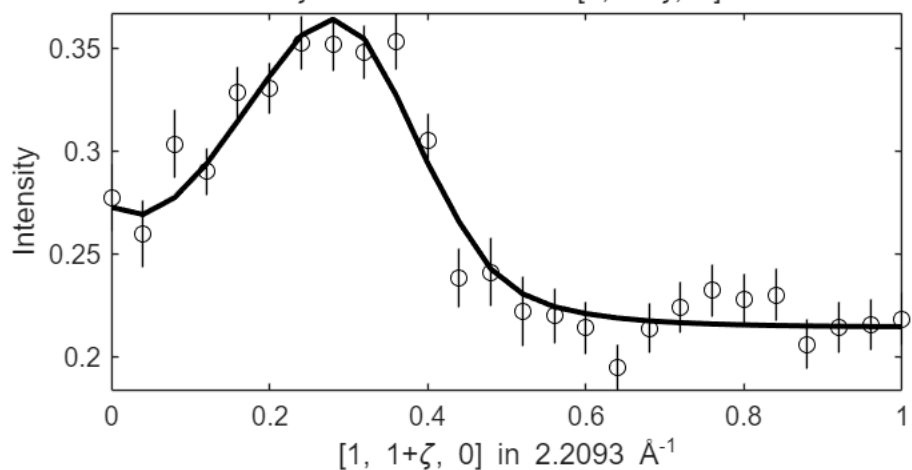


Fe_ei800, w2_800om1m1.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $88 \leq E \leq 100$ eV
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$

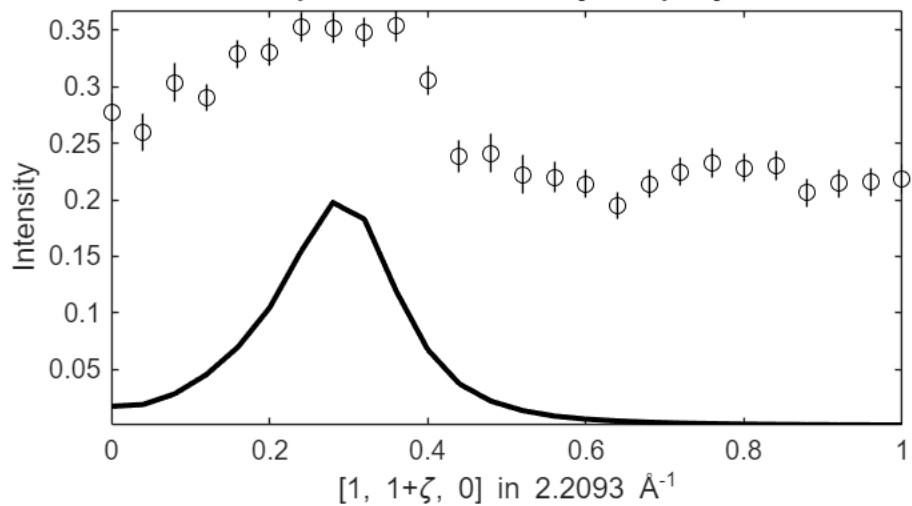


 **** En = 103±5 going to 293

Fe_ei800, w2_800om1m1.sqw
 S= 1 ± 0.042 ; J0= 35 ± 1.2 ; J1= 0 ± 0 ; gamma=70.6402±2.28561
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $98 \leq E \leq 100$ eV
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$

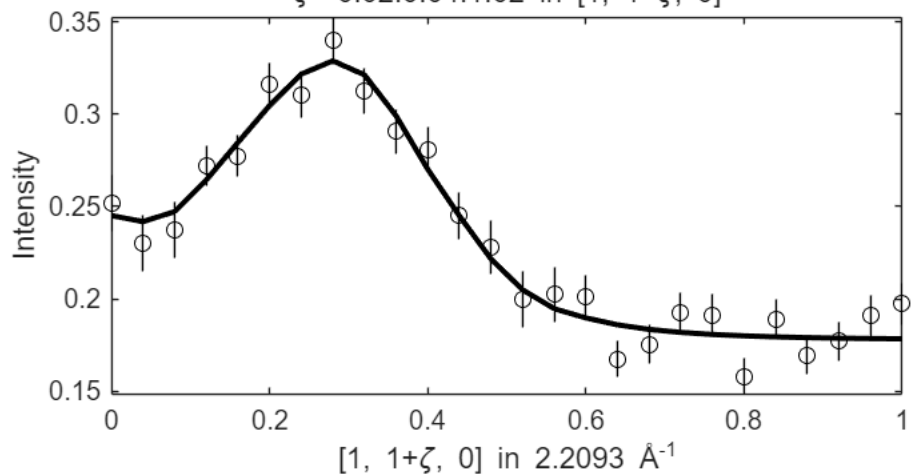


Fe_ei800, w2_800om1m1.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $98 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$

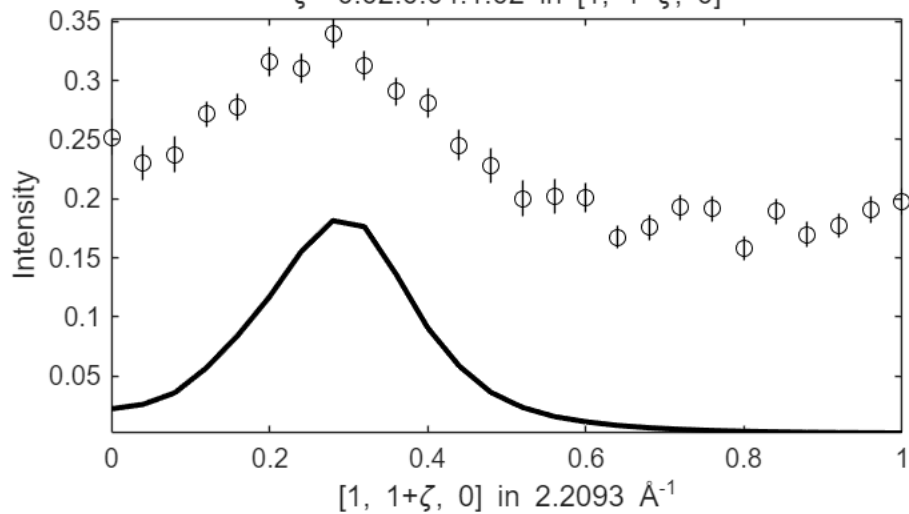


 **** En = 113±5 going to 293

Fe_ei800, w2_800om1m1.sqw
 $S = 1.9 \pm 0.073$; $J_0 = 47 \pm 1.5$; $J_1 = 0 \pm 0$; $\gamma = 165.495 \pm 4.38887$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $108 \leq E \leq$
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$

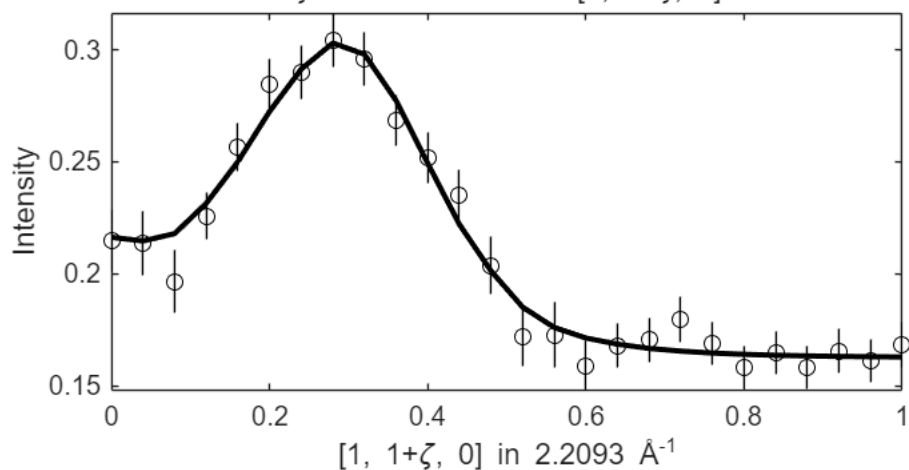


Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $108 \leq E \leq$
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



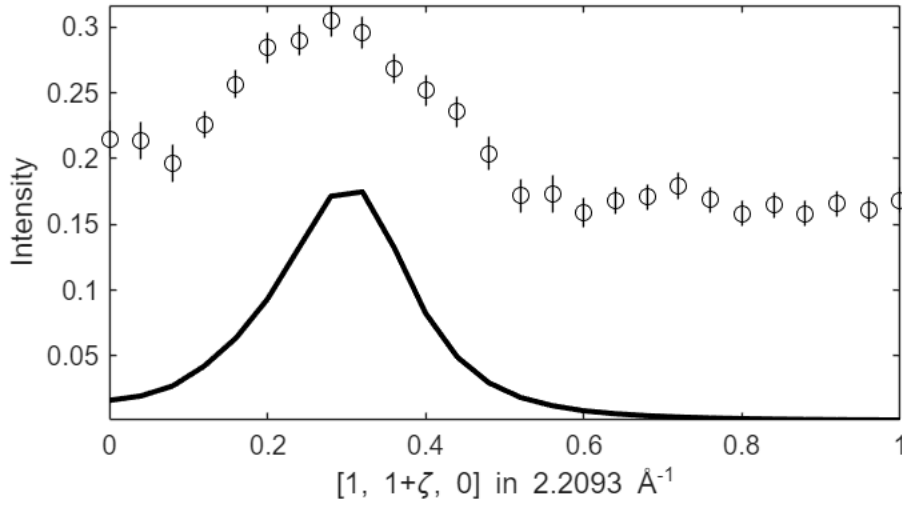
 **** En = 123±5 going to 293

Fe_ei800, w2_800om1m1.sqw
 $S = 1.3 \pm 0.035$; $J_0 = 41 \pm 0.86$; $J_1 = 0 \pm 0$; $\gamma = 100.415 \pm 2.00658$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $118 \leq E \leq$
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



Fe_ei800, w2_800om1m1.sqw

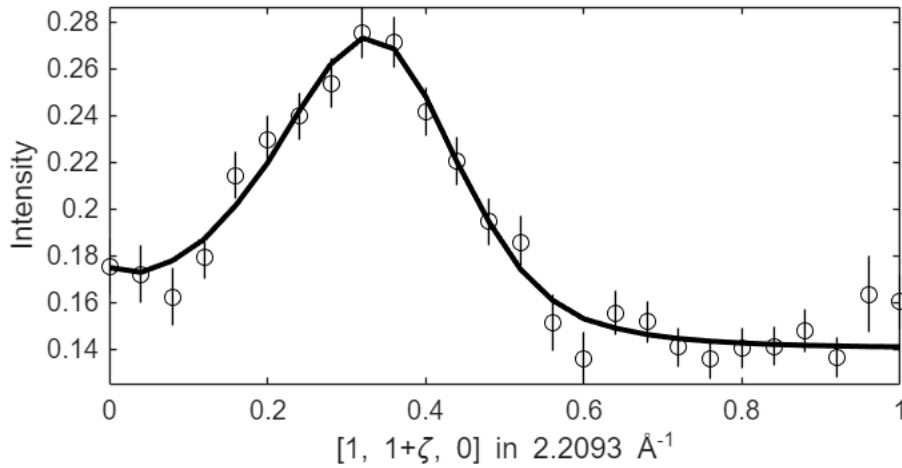
$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $118 \leq E \leq$
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



 **** En = 133 ± 5 going to 293

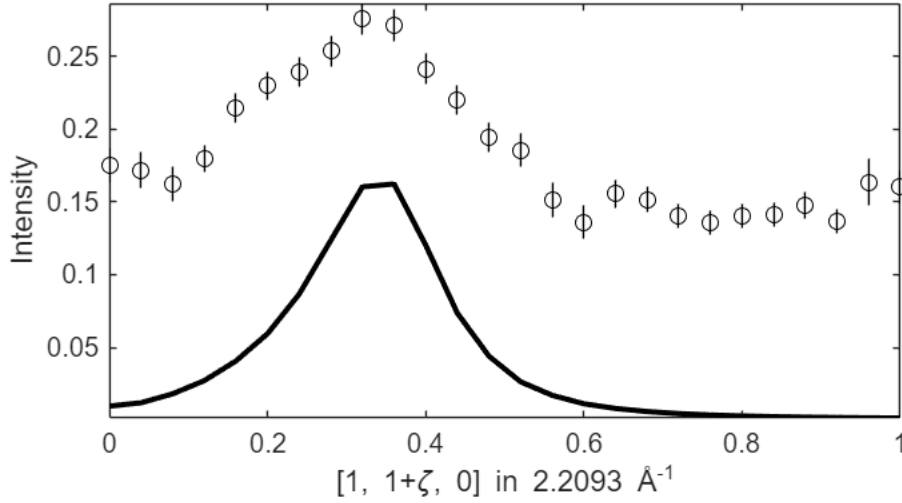
Fe_ei800, w2_800om1m1.sqw

$S = 1.1 \pm 0.036$; $J_0 = 34 \pm 0.82$; $J_1 = 0 \pm 0$; $\gamma = 80.7784 \pm 1.89828$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $128 \leq E \leq$
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



Fe_ei800, w2_800om1m1.sqw

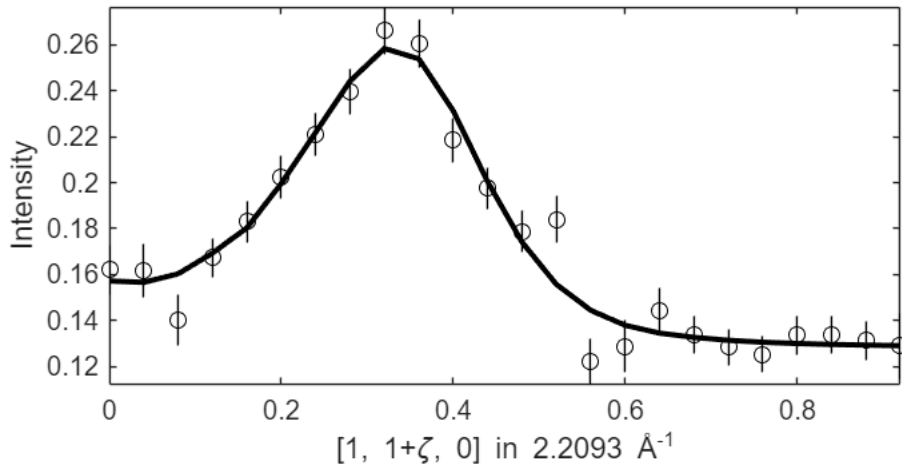
$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $128 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



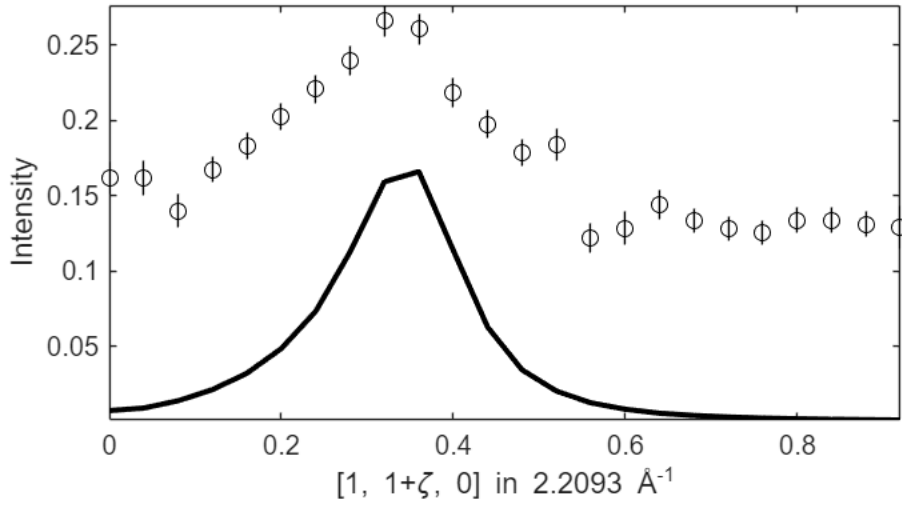
 **** En = 143 ± 5 going to 293

Fe_ei800, w2_800om1m1.sqw

$S = 0.9 \pm 0.036$; $J_0 = 35 \pm 0.84$; $J_1 = 0 \pm 0$; $\gamma = 64.7974 \pm 1.92189$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $138 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$

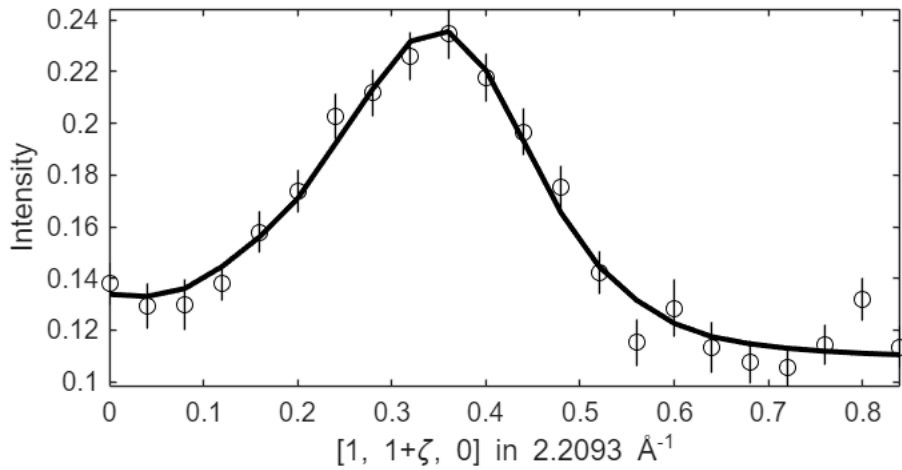


Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $138 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



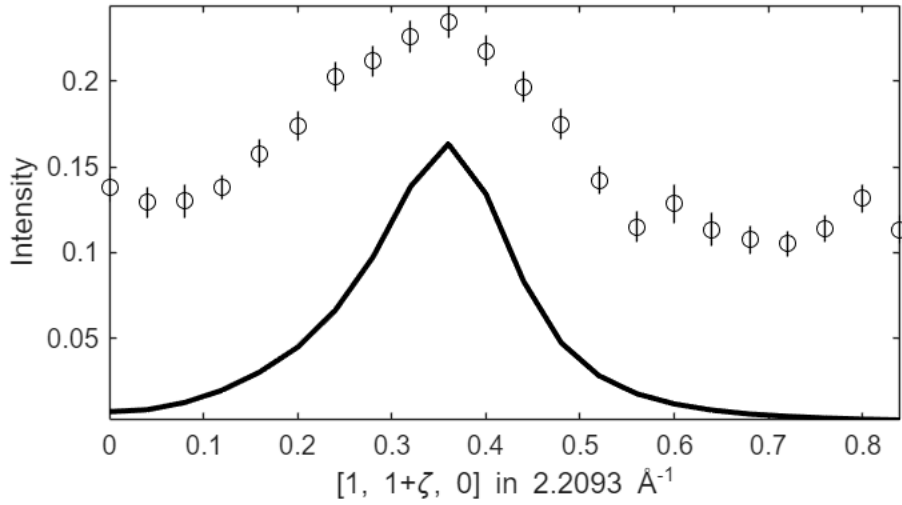
 **** En = 153±5 going to 293

Fe_ei800, w2_800om1m1.sqw
 S=0.99±0.038; J0= 35±0.74; J1= 0± 0; gamma=75.1782±1.78915
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $148 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



Fe_ei800, w2_800om1m1.sqw

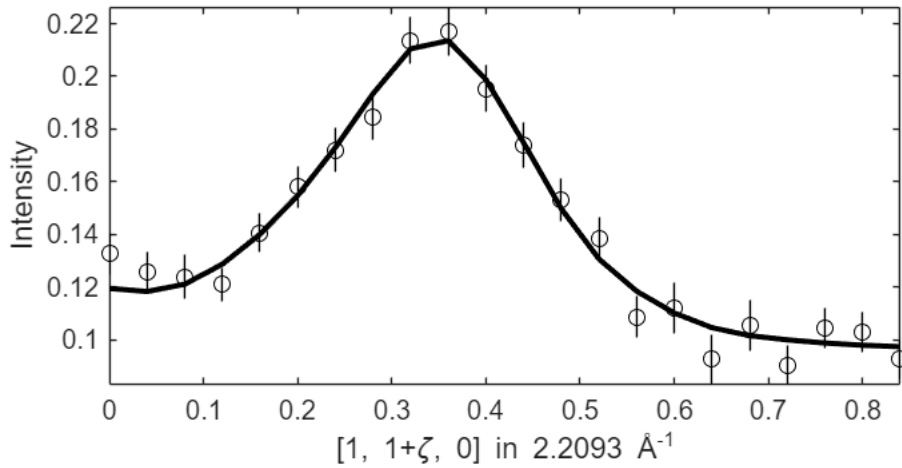
$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $148 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



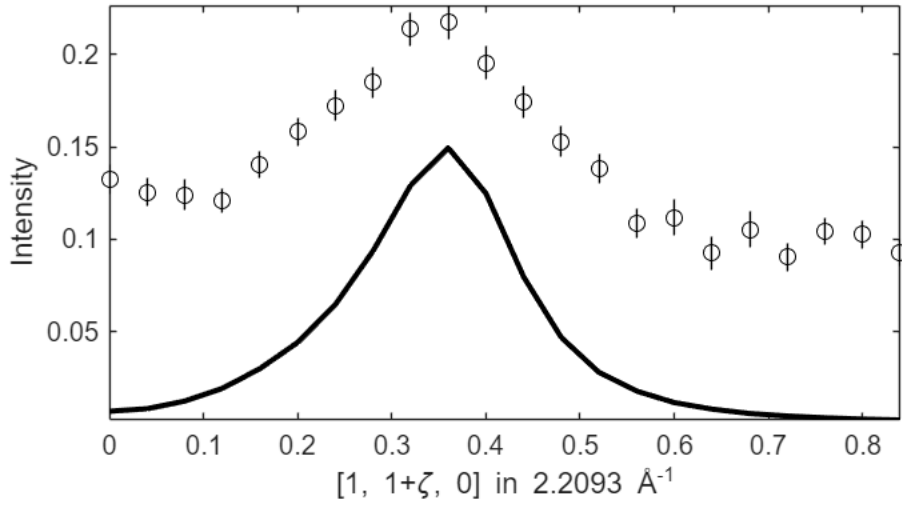
**** En = 163±5 going to 293

Fe_ei800, w2_800om1m1.sqw

S= 1 ± 0.037 ; J0= 37 ± 0.73 ; J1= 0 ± 0 ; gamma= 86.228 ± 1.83154
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $158 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$

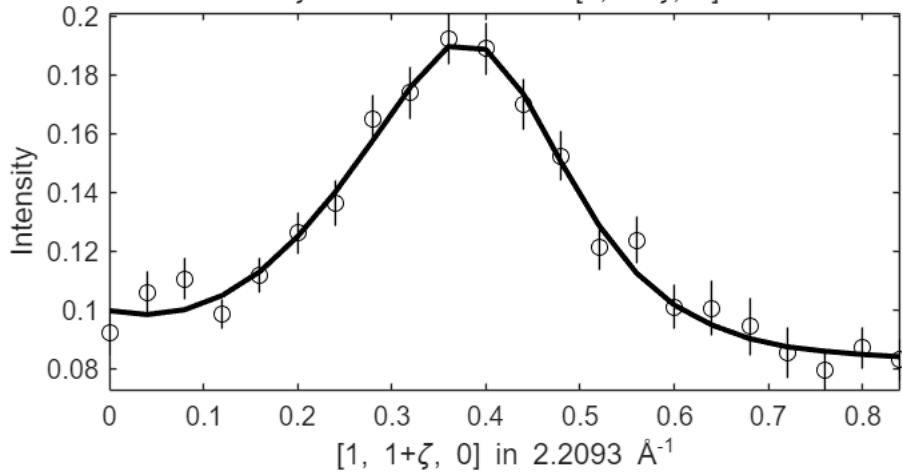


Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $158 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$

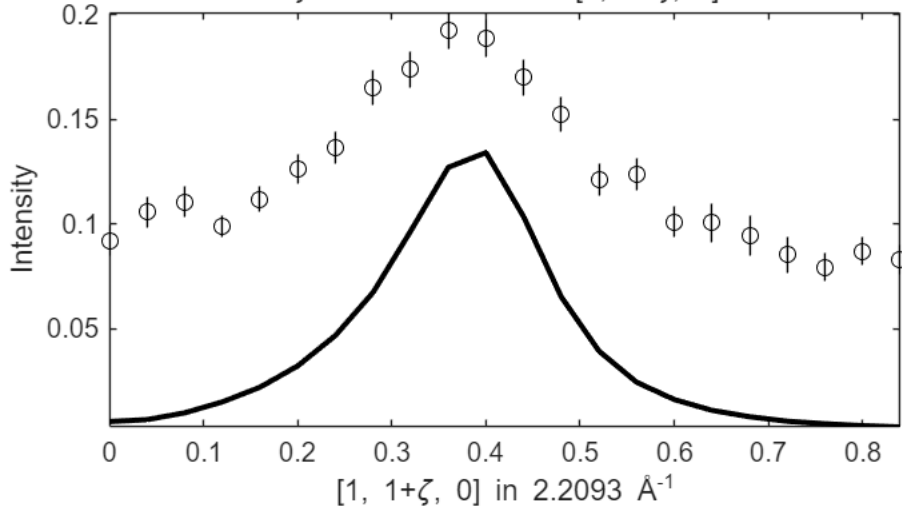


 **** En = 173±5 going to 293

Fe_ei800, w2_800om1m1.sqw
 $S=0.96\pm0.032$; $J0= 35\pm0.67$; $J1= 0\pm 0$; $\gamma=83.7801\pm2.08408$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $168 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$

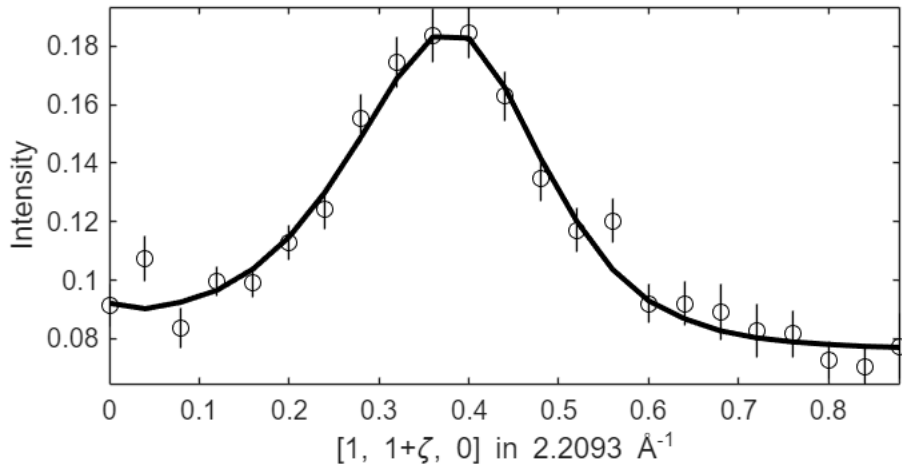


Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $168 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$

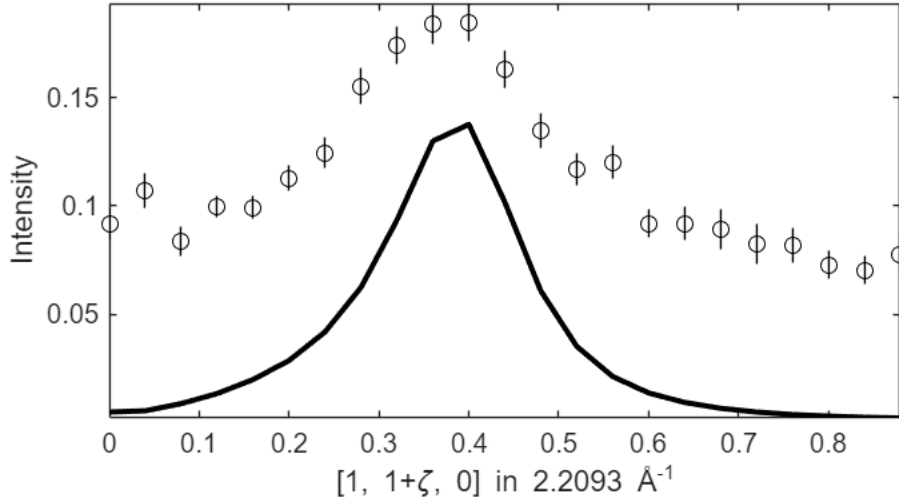


 **** En = 183±5 going to 293

Fe_ei800, w2_800om1m1.sqw
 $S=0.92\pm0.13$; $J_0= 36\pm 1$; $J_1= 0\pm 0$; $\gamma=77.0617\pm15.2532$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $178 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$

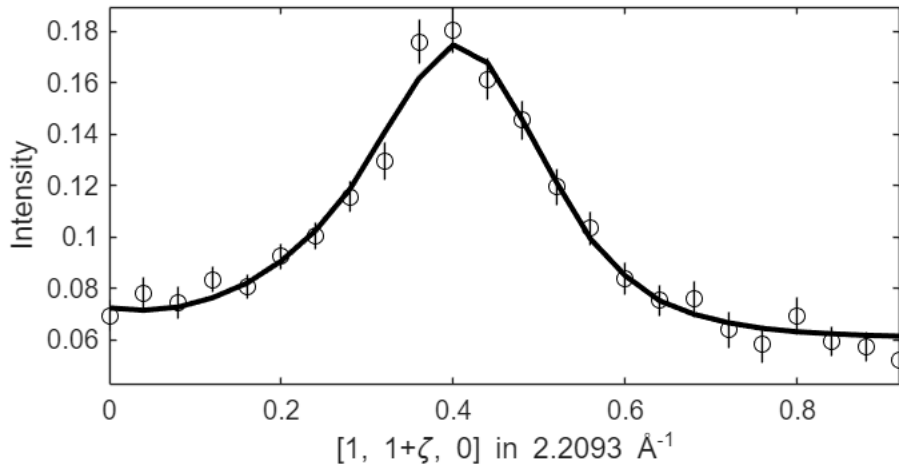


Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $178 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



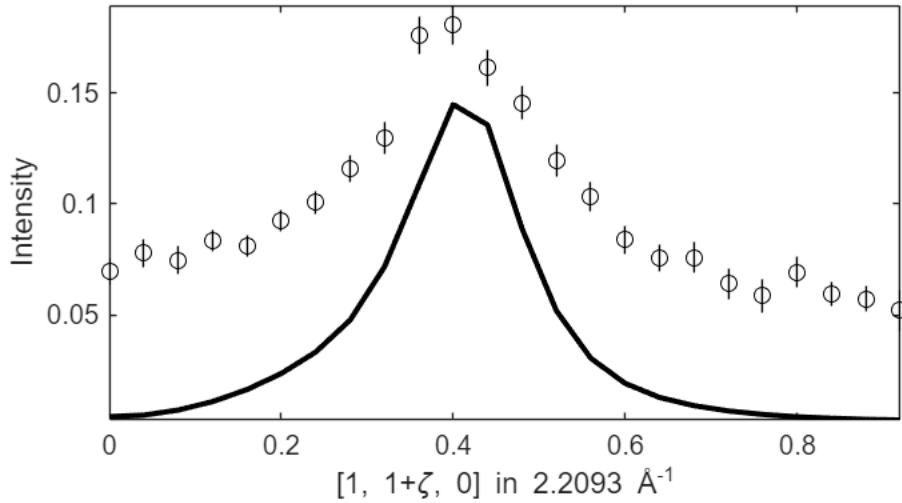
 **** En = 193±5 going to 293

Fe_ei800, w2_800om1m1.sqw
 $S=0.93\pm0.033$; $J_0= 34\pm0.53$; $J_1= 0\pm 0$; $\gamma=70.9711\pm1.47422$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $188 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



Fe_ei800, w2_800om1m1.sqw

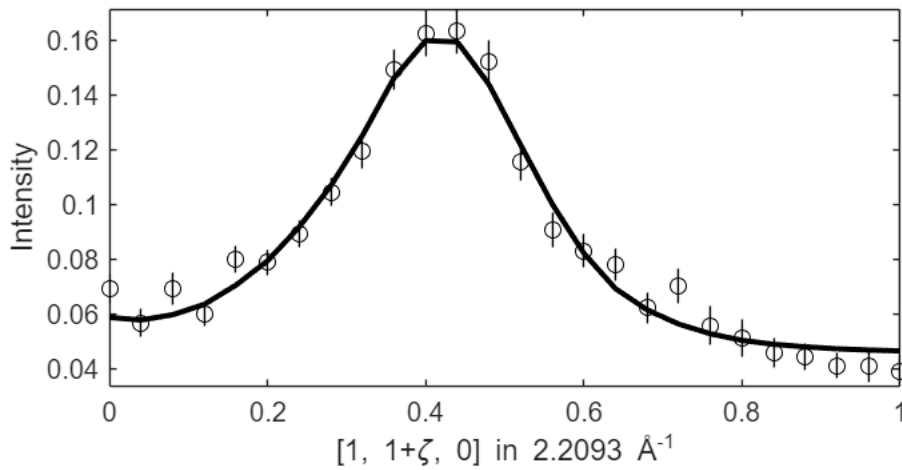
$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $188 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



**** En = 203±5 going to 293

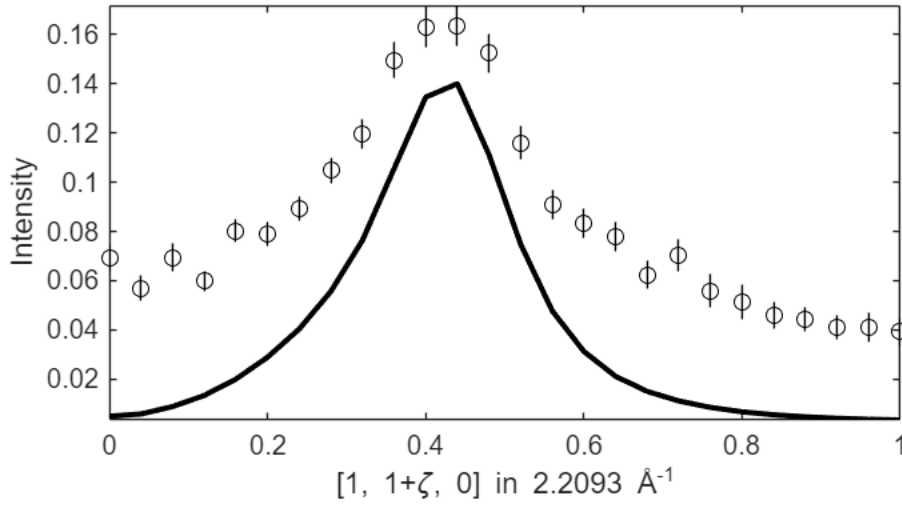
Fe_ei800, w2_800om1m1.sqw

S= 1.2±0.051; J0= 35±0.58; J1= 0± 0; gamma=95.9958±2.39682
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $198 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



Fe_ei800, w2_800om1m1.sqw

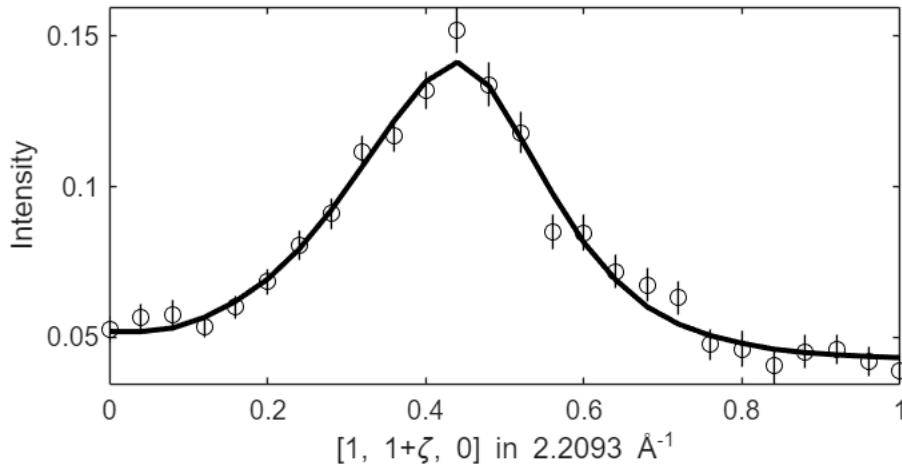
$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $198 \leq E \leq$:
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



 **** En = 213±5 going to 293

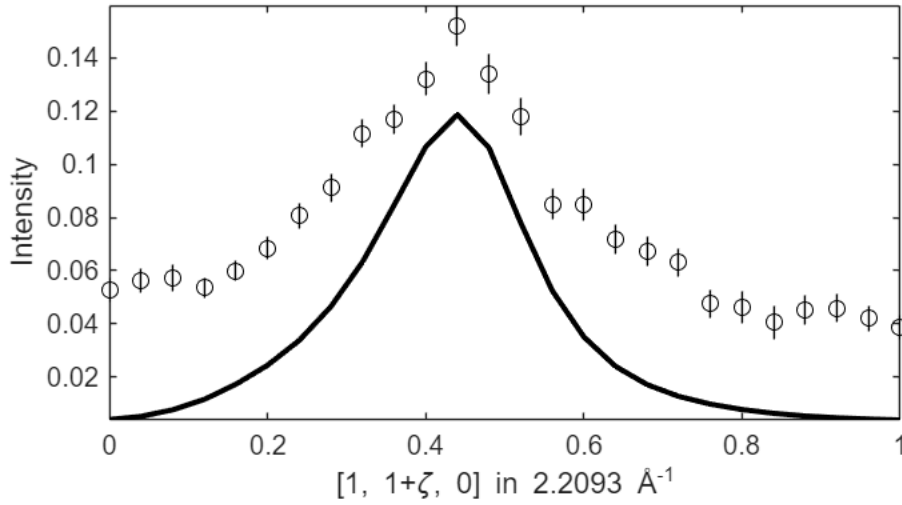
Fe_ei800, w2_800om1m1.sqw

S= 1.2 ± 0.12 ; J0= 35 ± 0.85 ; J1= 0 ± 0 ; gamma= 108.395 ± 15.4252
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $208 \leq E \leq$:
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



Fe_ei800, w2_800om1m1.sqw

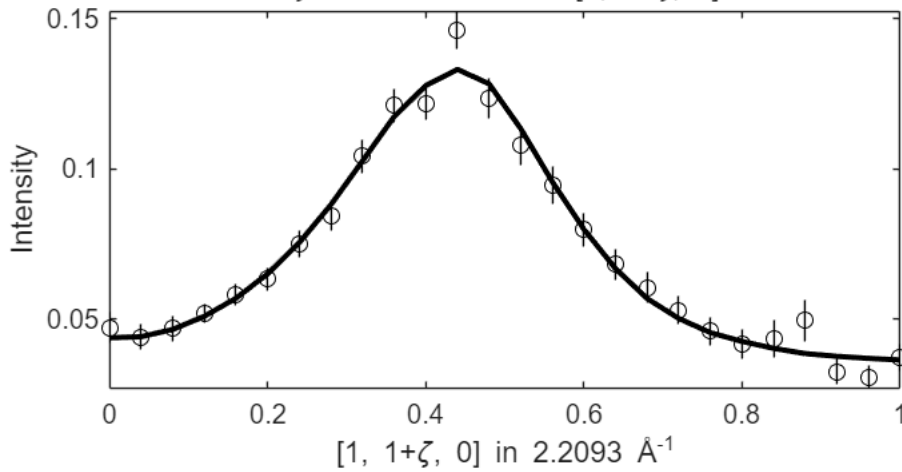
$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $208 \leq E \leq :$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



 **** En = 223±5 going to 293

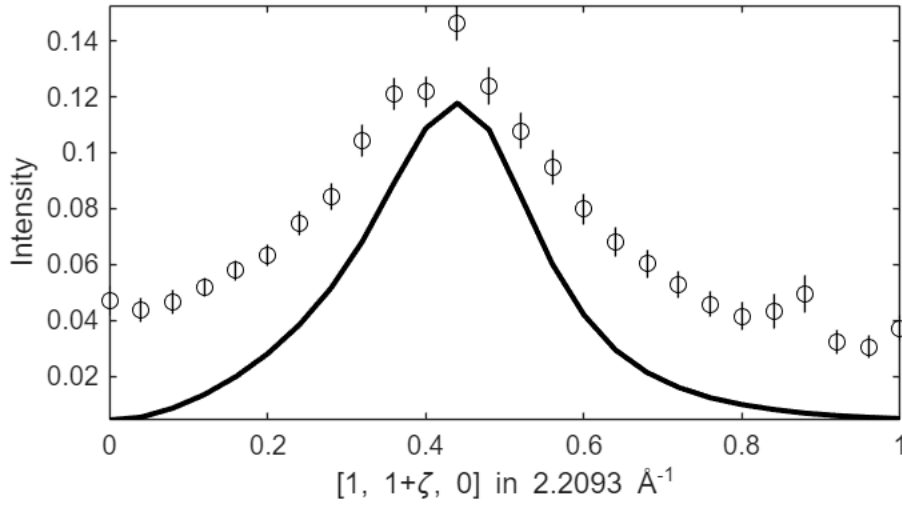
Fe_ei800, w2_800om1m1.sqw

S= 1.4 ± 0.15 ; J0= 37 ± 0.98 ; J1= 0 ± 0 ; gamma= 134.249 ± 18.0911
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $218 \leq E \leq :$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



Fe_ei800, w2_800om1m1.sqw

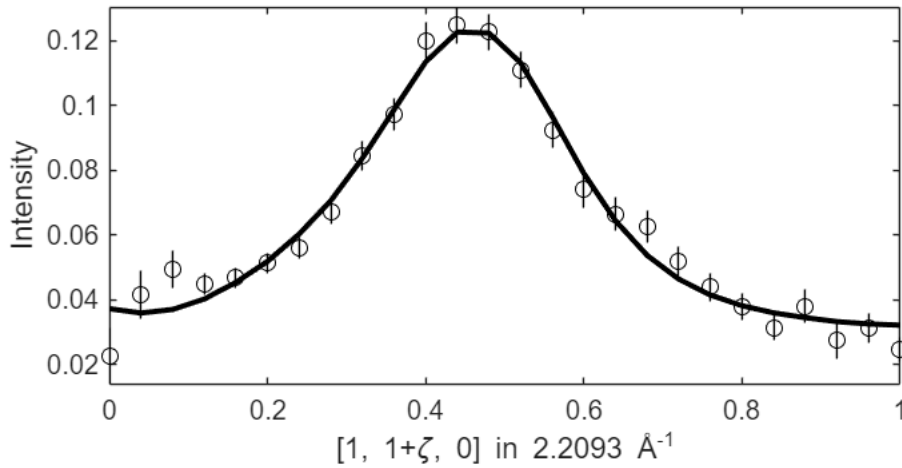
$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $218 \leq E \leq$;
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



 **** En = 233±5 going to 293

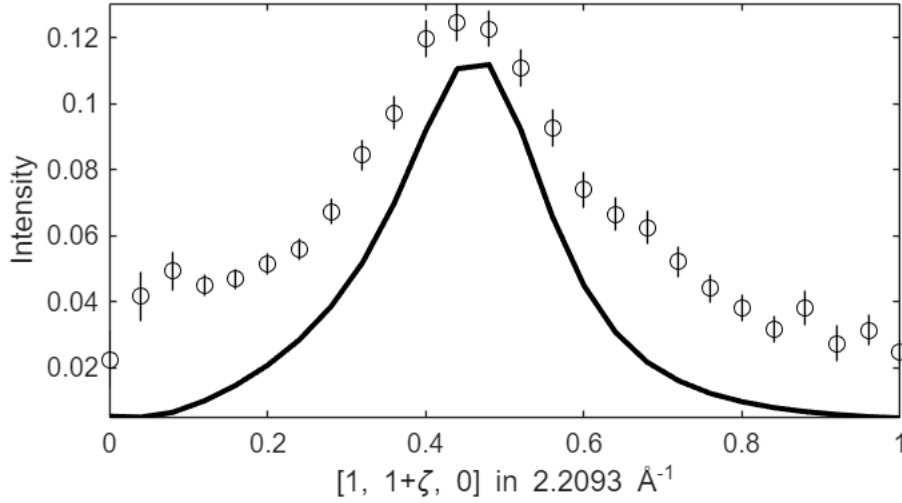
Fe_ei800, w2_800om1m1.sqw

S= 1.2 ± 0.061 ; J0= 35 ± 0.51 ; J1= 0 ± 0 ; gamma= 111.856 ± 3.52213
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $228 \leq E \leq$;
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



Fe_ei800, w2_800om1m1.sqw

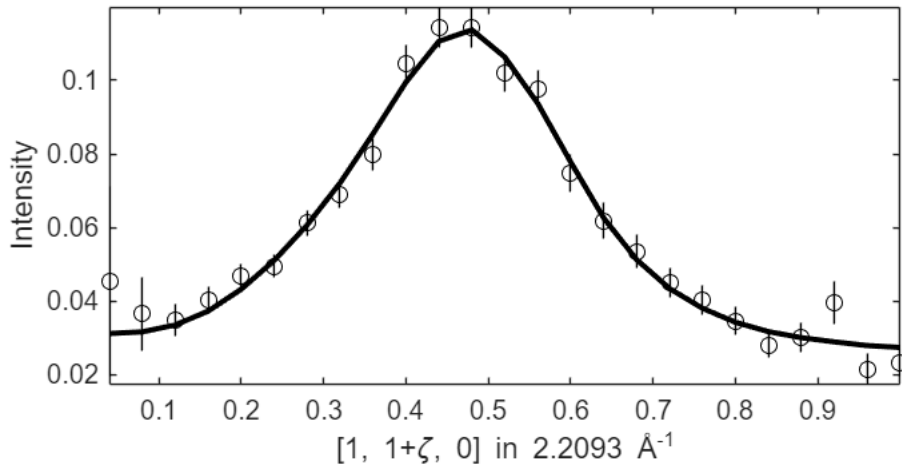
$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $228 \leq E \leq$:
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



 **** En = 243±5 going to 293

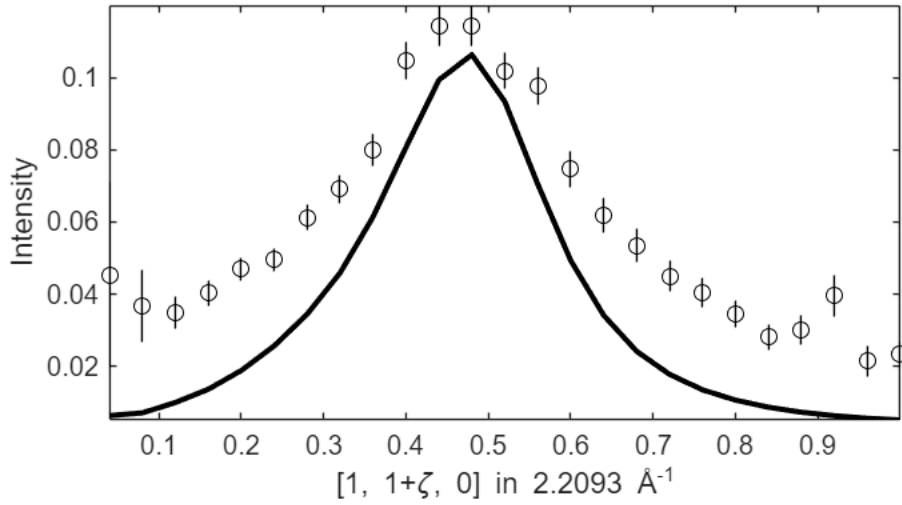
Fe_ei800, w2_800om1m1.sqw

S= 1.2±0.11; J0= 35± 0.6; J1= 0± 0; gamma=115.787±13.7022
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $238 \leq E \leq$:
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



Fe_ei800, w2_800om1m1.sqw

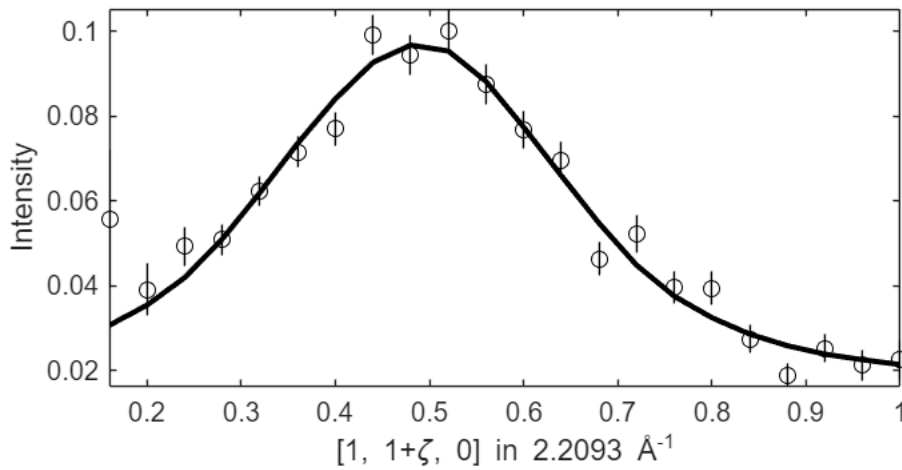
$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $238 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



 **** En = 253±5 going to 293

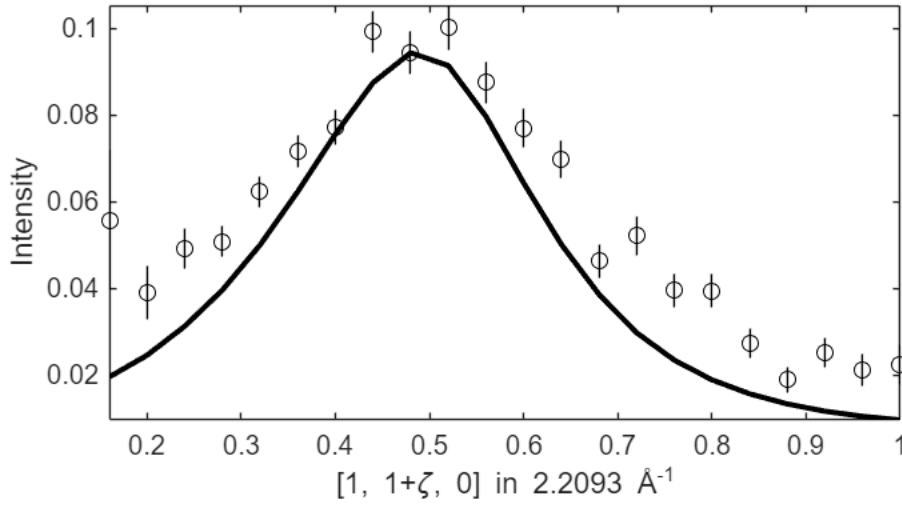
Fe_ei800, w2_800om1m1.sqw

S= 1.6 ± 0.34 ; J0= 37 ± 1.4 ; J1= 0 ± 0 ; gamma= 183.926 ± 38.4963
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $248 \leq E \leq$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



Fe_ei800, w2_800om1m1.sqw

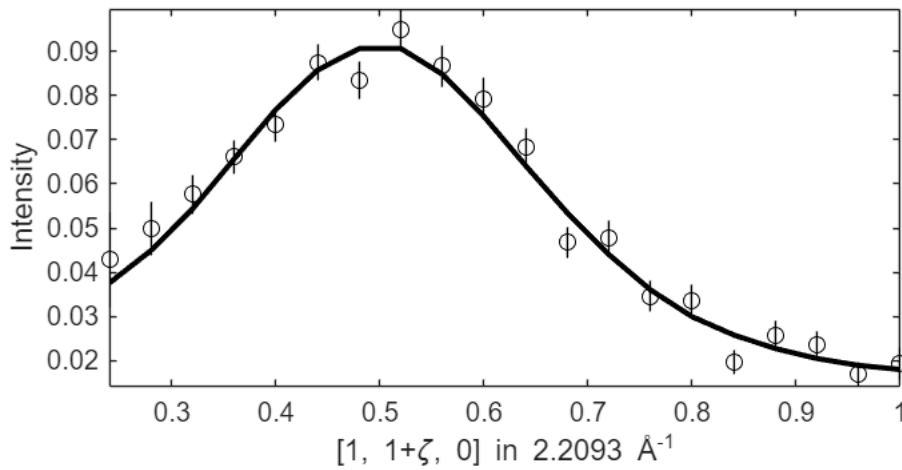
$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $248 \leq E \leq :$
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



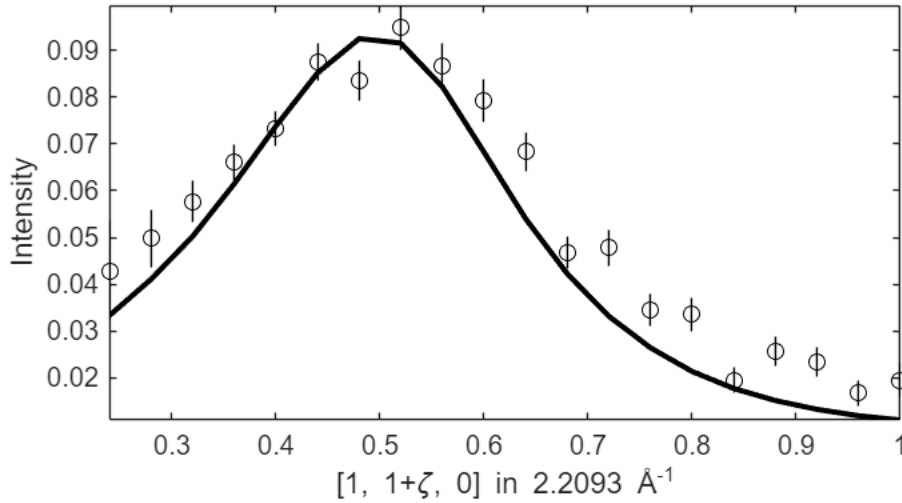
 **** En = 263±5 going to 293

Fe_ei800, w2_800om1m1.sqw

$S = 1.7 \pm 0.4$; $J_0 = 37 \pm 2.2$; $J_1 = 0 \pm 0$; $\gamma = 198.277 \pm 51.8342$
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $258 \leq E \leq :$
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$

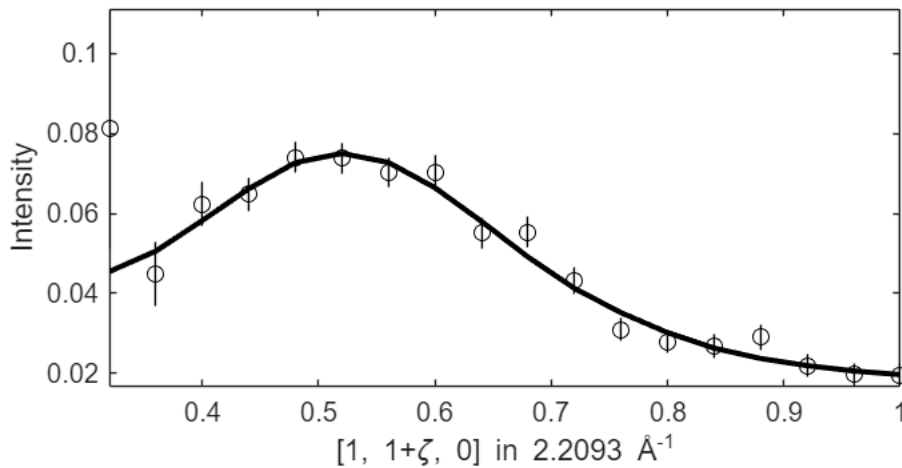


Fe_ei800, w2_800om1m1.sqw
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $258 \leq E \leq :$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



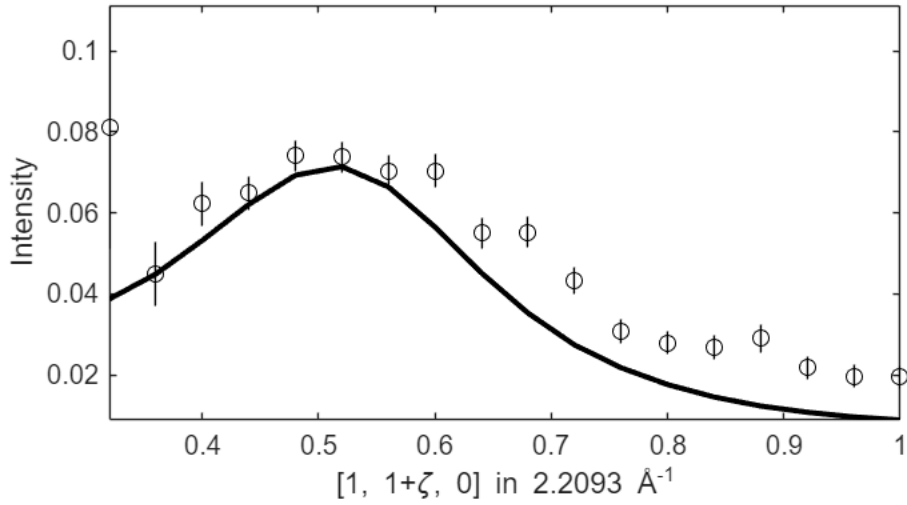
 **** En = 273±5 going to 293

Fe_ei800, w2_800om1m1.sqw
 S= 1.3±0.22; J0= 36±0.92; J1= 0± 0; gamma=181.171±29.4231
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $268 \leq E \leq :$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



Fe_ei800, w2_800om1m1.sqw

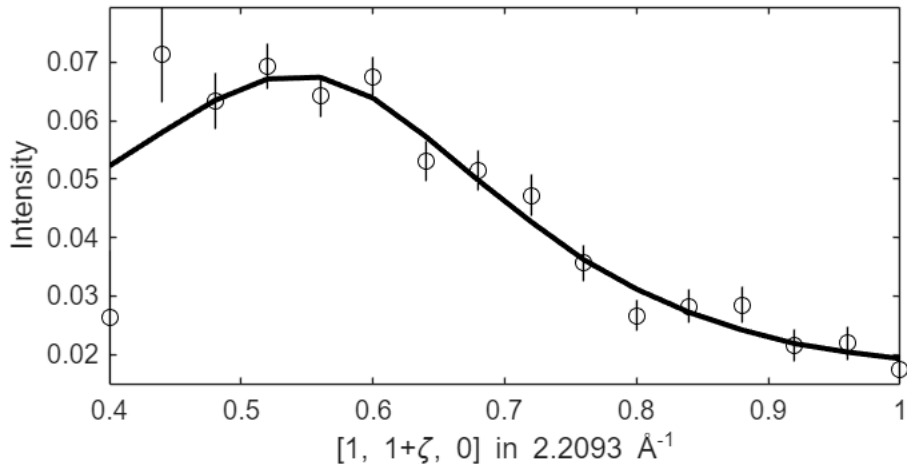
$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $268 \leq E \leq :$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



 **** En = 283±5 going to 293

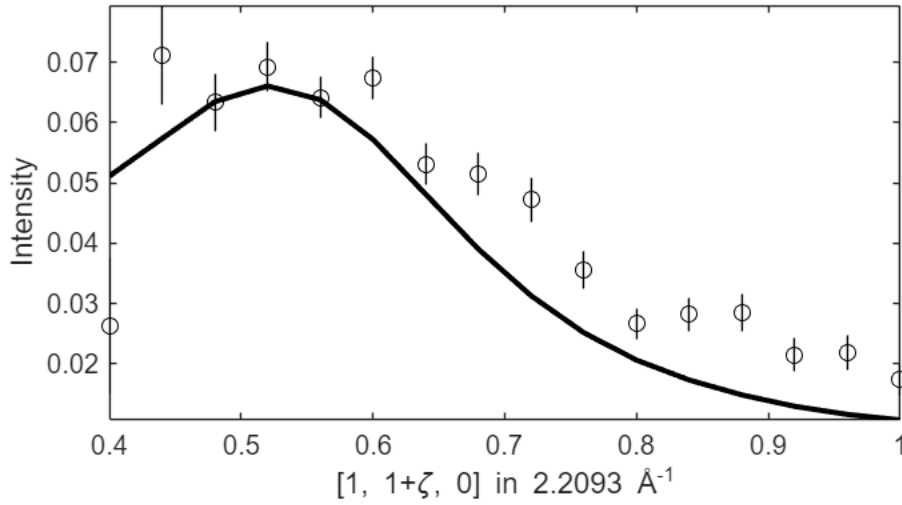
Fe_ei800, w2_800om1m1.sqw

S= 1.3± 0.5; J0= 36± 2.7; J1= 0± 0; gamma=205.087±75.1305
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $278 \leq E \leq :$
 $\zeta=-0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



Fe_ei800, w2_800om1m1.sqw

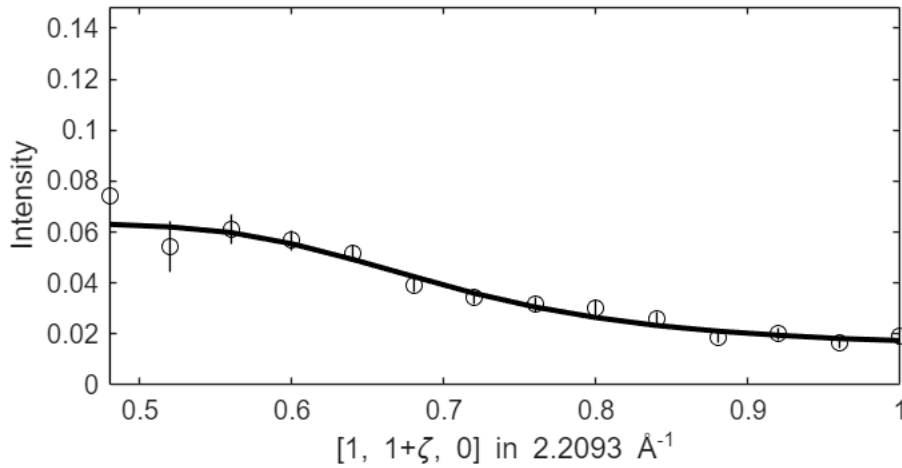
$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $278 \leq E \leq :$
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



 **** En = 293±5 going to 293

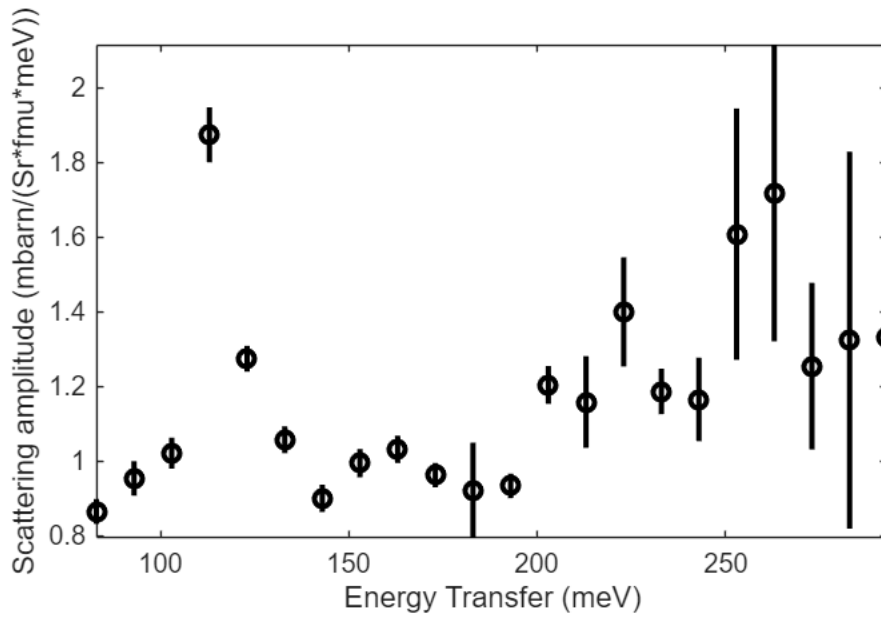
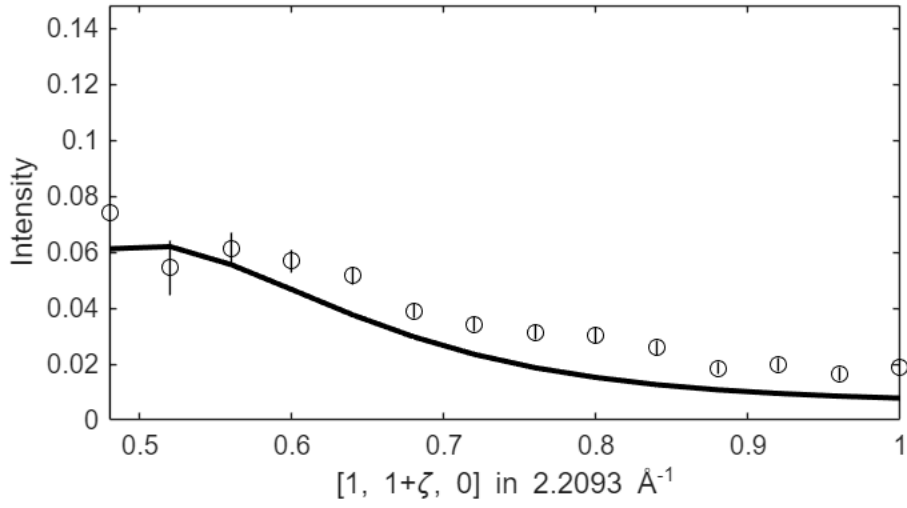
Fe_ei800, w2_800om1m1.sqw

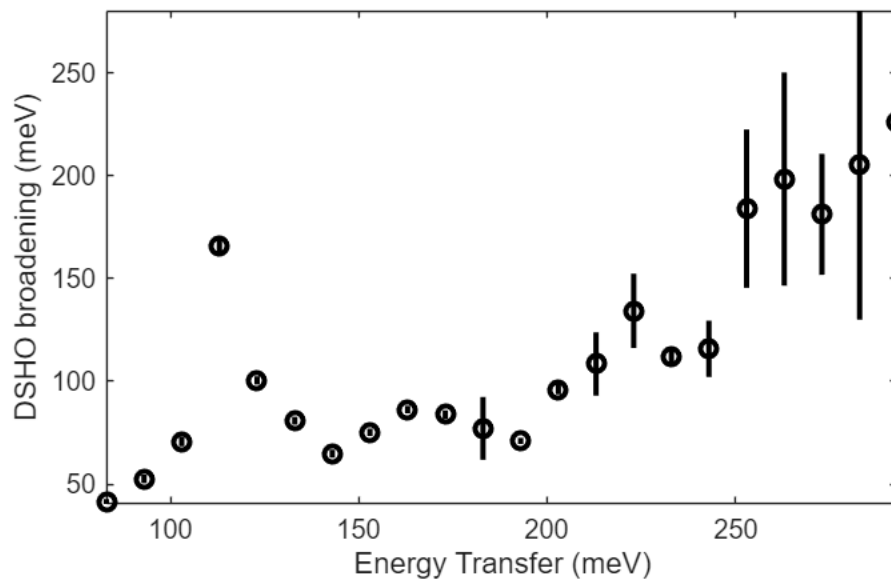
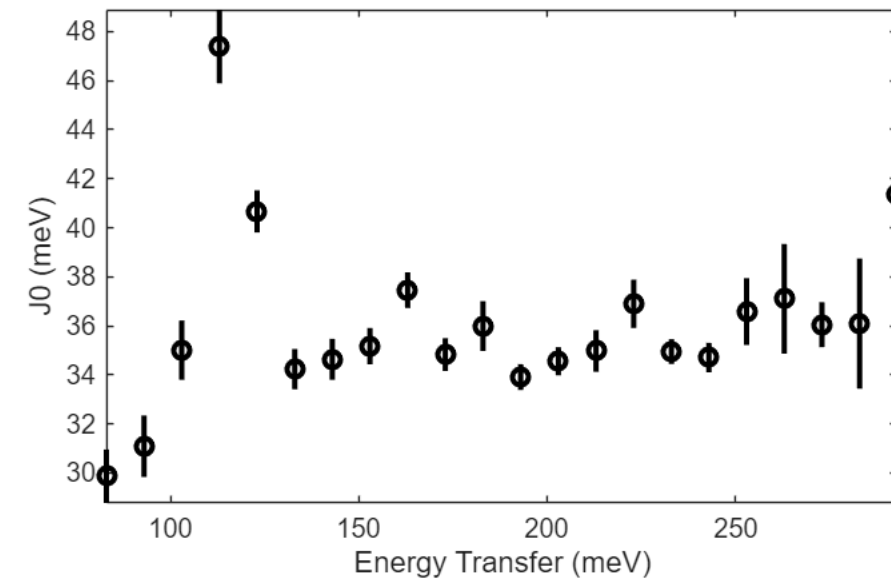
S= 1.3 ± 0.21 ; J0= 41 ± 0.98 ; J1= 0 ± 0 ; gamma=226.277±47.4896
 $0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $288 \leq E \leq :$
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$



Fe_ei800, w2_800om1m1.sqw

$0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $288 \leq E \leq$:
 $\zeta = -0.02:0.04:1.02$ in $[1, 1+\zeta, 0]$





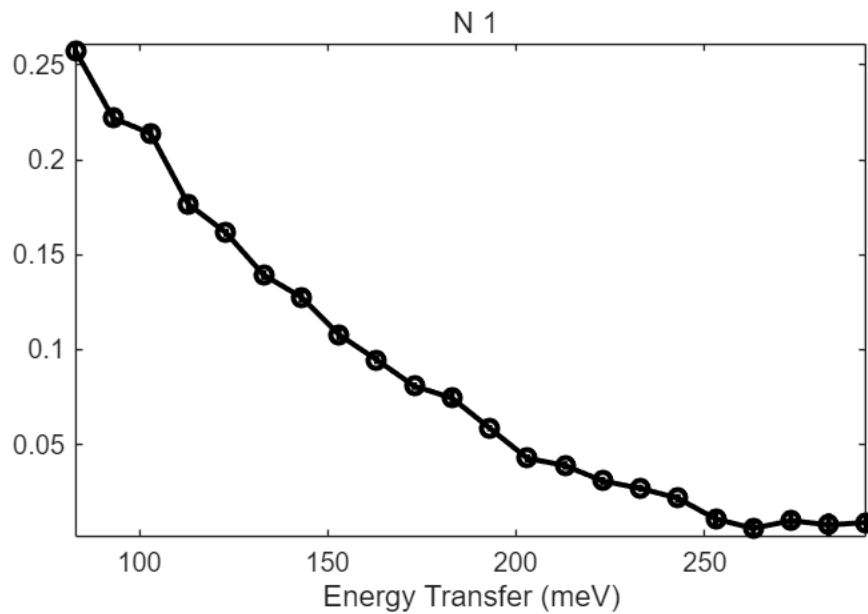
```

c1 = GH_FFCutsFits.(selectCuts);
GH_FFCutsFits.(selectCuts) = [c1{:}];

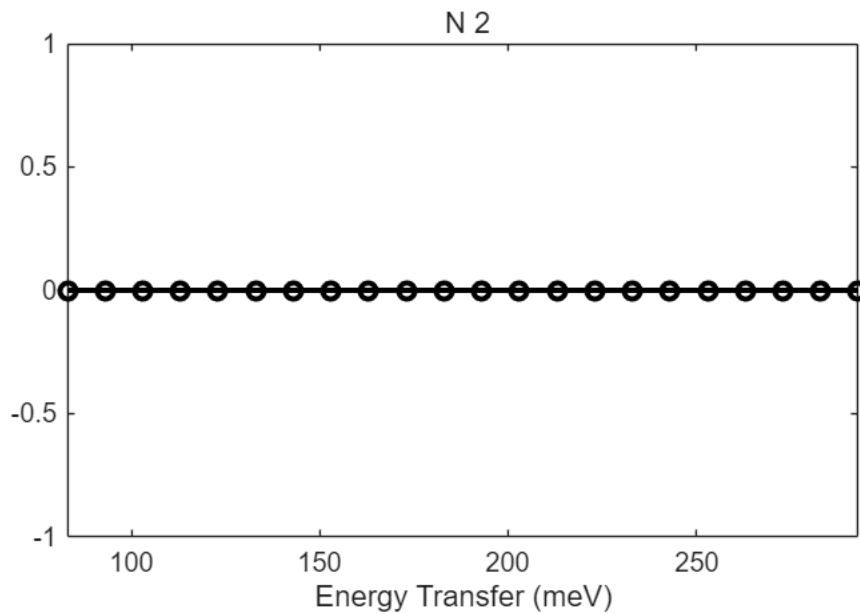
beep
beep
beep

all_fit_par = GH_FFCutsFits.([selectCuts,'_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GH.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GH_',selectCuts,'.pdf']);
export(thisFile,targ_file);
```

```
fit_src_struct = GH_FFCutsFits;  
save(CutFitFileGH,"fit_src_struct",-v7.3);  
close all;  
%end
```